

ACCOUNTING FOR MANAGERS

Complete Unit-Wise Study Notes

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Unit I: Management Accounting & Financial Statement Analysis

Unit II: Ratio Analysis & Fund/Cash Flow Statements

Unit III: Business Budgeting, Marginal Costing & BEP

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UNIT I — MANAGEMENT ACCOUNTING & FINANCIAL STATEMENT ANALYSIS

1.1 MANAGEMENT ACCOUNTING – CONCEPT & MEANING

Definition: Management Accounting is the process of identification, measurement, accumulation, analysis, preparation, interpretation, and communication of financial information used by management to plan, evaluate, and control within an organisation and to assure appropriate use of and accountability for its resources.

Simple Meaning: Management Accounting provides financial and non-financial data to managers so they can make better decisions. Unlike Financial Accounting, it is *not compulsory* and is used *internally* by management.

1.2 CHARACTERISTICS OF MANAGEMENT ACCOUNTING

Characteristic	Explanation
Selective in Nature	It selects only relevant data needed for a specific managerial decision.
Future-Oriented	It focuses on planning and forecasting future events, not just recording past transactions.
No Fixed Format	There is no prescribed format. Reports are prepared as per management needs.
Combination of Sciences	It integrates financial accounting, cost accounting, statistics, economics, and management.
Cause and Effect Analysis	It analyses why profits/losses occurred, not just that they occurred.
Flexible	It can be changed and adapted to meet management requirements.
Not Legally Compulsory	Management accounting is optional; no statutory requirement to maintain it.
Uses Both Monetary & Non-Monetary Data	It considers quantitative as well as qualitative information.

1.3 DIFFERENCES: FINANCIAL vs MANAGEMENT vs COST ACCOUNTING

Basis	Financial Accounting	Management Accounting	Cost Accounting
Purpose	Report to external parties	Aid internal decision-making	Ascertain cost of product/service
Users	Shareholders, creditors, govt	Managers & executives	Management & cost controllers

Basis	Financial Accounting	Management Accounting	Cost Accounting
Legal Requirement	Compulsory (Companies Act)	Not compulsory	Required in some industries
Time Focus	Past (historical)	Future (forward-looking)	Past & Future both
Format	Prescribed (GAAP/Ind AS)	No fixed format	Specific cost formats
Scope	Whole organisation	Specific departments/decisions	Production & cost centres
Data Type	Monetary only	Monetary + Non-monetary	Monetary (cost data)
Reports	Annual reports	Daily/weekly/monthly reports	Cost sheets, cost statements
Audit	Statutory audit required	No audit required	Internal audit usually

■ **EXAM TIP: In exams, differences tables fetch full marks. Always write 8–10 points with proper basis.**

1.4 TECHNIQUES OF MANAGEMENT ACCOUNTING

- ◆ Financial Statement Analysis – Ratio analysis, trend analysis, comparative statements
- ◆ Funds Flow & Cash Flow Analysis – Movement of funds and cash
- ◆ Marginal Costing – Cost-volume-profit analysis
- ◆ Budgetary Control – Planning and controlling through budgets
- ◆ Standard Costing – Setting standards and measuring variances
- ◆ Capital Budgeting – Long-term investment decisions
- ◆ Statistical Methods – Index numbers, probability, regression
- ◆ Reporting – Communicating financial data to management

Objectives of Management Accounting:

- Assist management in planning by providing financial data
- Help in controlling operations through budgets and standards
- Aid in decision-making by supplying relevant cost-benefit data
- Measure and evaluate performance of departments and managers
- Coordinate the activities of different departments
- Facilitate communication between management levels

Importance of Management Accounting:

- Helps in effective planning and forecasting
- Facilitates sound financial control
- Provides basis for important business decisions
- Helps in evaluating business performance
- Enables delegation of authority through budgeting
- Motivates employees through target-setting

1.5 MANAGEMENT ACCOUNTANT – DUTIES, STATUS, FUNCTIONS & RESPONSIBILITY

Definition: A Management Accountant (also called Controller or Comptroller) is a professional who prepares financial and accounting reports for use by internal management.

Area	Description
Duties	Maintaining management information systems, preparing budgets, cost reports, variance analysis, financial forecasts
Status	Senior position in organisational hierarchy; reports directly to CFO or top management; participates in policy decisions
Functions	Planning, Controlling, Decision Making, Performance Measurement, Risk Assessment, Internal Audit coordination
Responsibility	Accuracy of reports, confidentiality of data, compliance with accounting standards, providing unbiased recommendations

1.6 FINANCIAL STATEMENT ANALYSIS

Meaning: Financial Statement Analysis is the process of critically examining financial data contained in financial statements with the objective of making economic decisions. It involves reviewing balance sheets, income statements, and cash flow statements.

Objectives:

- To assess the profitability of the business
- To evaluate the short-term and long-term solvency
- To compare performance over time and with competitors
- To help investors make investment decisions
- To assist creditors in credit evaluation
- To judge management efficiency

Parties Interested in Financial Statements:

Party	Interest
Shareholders/Investors	To assess return on investment and safety of funds
Creditors & Lenders	To evaluate ability to repay loans and interest
Management	To plan, control and take corrective actions
Government & Tax Authorities	For taxation and regulatory compliance
Employees	To assess job security and wage-payment capacity
Customers	To evaluate long-term stability of suppliers
Research Analysts	To make recommendations on stocks and bonds

Types of Financial Analysis:

(A) Horizontal Analysis (Comparative Statement Analysis)

Meaning: Comparison of financial data *across multiple periods* (year over year). Shows absolute change and percentage change.

Formula:

$\text{Change in Amount} = \text{Current Year Figure} - \text{Base Year Figure}$

$\% \text{ Change} = (\text{Change in Amount} / \text{Base Year Figure}) \times 100$

— Horizontal Analysis Example

Particulars	2022 (Rs.)	2023 (Rs.)	Change (Rs.)	% Change
Sales	5,00,000	6,00,000	+1,00,000	+20%
Net Profit	50,000	75,000	+25,000	+50%
Total Assets	8,00,000	9,50,000	+1,50,000	+18.75%
Current Liabilities	1,20,000	1,00,000	-20,000	-16.67%

(B) Vertical Analysis (Common Size Statement)

Meaning: Each item in the financial statement is expressed as a percentage of a *base figure*. For Income Statement: Base = Net Sales. For Balance Sheet: Base = Total Assets.

$\text{Common Size \%} = (\text{Individual Item} / \text{Base Figure}) \times 100$

— Vertical Analysis Example – Common Size Income Statement

Particulars	Amount (Rs.)	% of Net Sales
Net Sales	5,00,000	100%
Cost of Goods Sold	3,00,000	60%
Gross Profit	2,00,000	40%
Operating Expenses	80,000	16%
Net Profit	1,20,000	24%

(C) Trend Analysis

Meaning: Calculation of trend ratios over multiple years by taking *one base year* and expressing subsequent years as percentages relative to it. Helps identify long-term patterns.

$\text{Trend \%} = (\text{Current Year Figure} / \text{Base Year Figure}) \times 100$

— Trend Analysis Example (Base Year = 2020)

Year	Sales (Rs.)	Trend %
2020 (Base)	4,00,000	100%
2021	4,40,000	110%
2022	5,00,000	125%
2023	5,60,000	140%
2024	6,40,000	160%

■ EXAM TIP: Trend analysis over 5+ years reveals whether the business is growing, stable, or declining. Upward trend = positive sign.

UNIT II — RATIO ANALYSIS & FUND/CASH FLOW STATEMENTS

2.1 RATIO ANALYSIS – MEANING & UTILITY

Meaning: A Ratio is a mathematical relationship between two related figures. **Ratio Analysis** is the process of computing, determining and presenting the relationship of items and groups of items in financial statements.

Utility/Advantages of Ratio Analysis:

- Simplifies financial statements into understandable form
- Helps in inter-firm and intra-firm comparison
- Assists in forecasting and budgeting
- Identifies problem areas requiring management attention
- Measures liquidity, profitability, and solvency
- Useful for investors, creditors, and management

2.2 PROFITABILITY RATIOS

Profitability Ratios measure the ability of a business to generate **profit relative to sales, assets, or equity**.

Ratio Name	Formula	Significance
Gross Profit Ratio	$GP / \text{Net Sales} \times 100$	Measures manufacturing/trading efficiency. Higher = better.
Net Profit Ratio	$\text{Net Profit} / \text{Net Sales} \times 100$	Overall profitability after all expenses. Higher = better.
Operating Profit Ratio	$\text{Operating Profit} / \text{Net Sales} \times 100$	Profit from core operations. Excludes non-operating income.
Return on Capital Employed (ROCE)	$\text{Net Profit (after tax)} / \text{Capital Employed} \times 100$	Efficiency of overall capital use. Ideal: >15%
Return on Equity (ROE)	$\text{Net Profit after tax} / \text{Shareholders' Equity} \times 100$	Returns to shareholders. Ideal: >12–15%
Return on Assets (ROA)	$\text{Net Profit} / \text{Total Assets} \times 100$	How efficiently assets generate profit.
Earnings Per Share (EPS)	$\text{Net Profit after tax} / \text{No. of Equity Shares}$	Profit per share. Important for investors.

— Solved Example – Profitability Ratios

Given Data: Net Sales = Rs.10,00,000 | COGS = Rs.6,00,000 | Operating Expenses = Rs.1,50,000
 | Net Profit = Rs.1,00,000 | Capital Employed = Rs.8,00,000 | Equity = Rs.5,00,000

Item	Calculation	Result
Gross Profit	$10,00,000 - 6,00,000 = \text{Rs.}4,00,000$	GP Ratio = $4,00,000 / 10,00,000 \times 100 = 40\%$
Operating Profit	$\text{GP} - \text{Op. Exp} = 4,00,000 - 1,50,000 = \text{Rs.}2,50,000$	Op. Profit Ratio = 25%
Net Profit Ratio	$1,00,000 / 10,00,000 \times 100$	= 10%
ROCE	$1,00,000 / 8,00,000 \times 100$	= 12.5%
ROE	$1,00,000 / 5,00,000 \times 100$	= 20%

2.3 ACTIVITY RATIOS (EFFICIENCY / TURNOVER RATIOS)

Activity Ratios measure how **efficiently a business uses its assets** to generate sales/revenue.
 Also called Turnover Ratios.

Ratio Name	Formula	Significance
Inventory / Stock Turnover Ratio	$\text{COGS} / \text{Average Stock (or Net Sales / Closing Stock)}$	How many times stock is sold per year. High ratio = efficient.
Debtors Turnover Ratio	$\text{Net Credit Sales} / \text{Average Debtors}$	Efficiency in collecting credit sales. High = better collection.
Creditors Turnover Ratio	$\text{Net Credit Purchases} / \text{Average Creditors}$	How quickly payables are cleared. Lower = more payment days.
Working Capital Turnover Ratio	$\text{Net Sales} / \text{Net Working Capital}$	Efficiency of working capital use in generating sales.
Fixed Asset Turnover Ratio	$\text{Net Sales} / \text{Net Fixed Assets}$	Revenue generated per rupee of fixed assets.
Total Asset Turnover Ratio	$\text{Net Sales} / \text{Total Assets}$	Overall asset utilisation efficiency.
Debtors Collection Period	$365 / \text{Debtors Turnover Ratio}$	Average days to collect from debtors. Lower = better.
Creditors Payment Period	$365 / \text{Creditors Turnover Ratio}$	Average days to pay creditors.

= Solved Example – Activity Ratios

Given: Net Sales = Rs.12,00,000 | COGS = Rs.8,00,000 | Opening Stock = Rs.80,000 | Closing Stock = Rs.1,20,000 | Debtors = Rs.1,50,000 | Working Capital = Rs.2,00,000

Item	Calculation	Result
Average Stock	$(80,000 + 1,20,000) / 2 = \text{Rs.}1,00,000$	

Item	Calculation	Result
Stock Turnover Ratio	$8,00,000 / 1,00,000$	= 8 times
Debtors Turnover Ratio	$12,00,000 / 1,50,000$	= 8 times
Debtors Collection Period	$365 / 8$	= ~45.6 days
Working Capital Turnover	$12,00,000 / 2,00,000$	= 6 times

2.4 FINANCIAL POSITION RATIOS (LIQUIDITY & SOLVENCY RATIOS)

Liquidity Ratios measure short-term ability to pay debts. **Solvency Ratios** measure long-term financial health.

Ratio Name	Formula	Significance
Current Ratio	Current Assets / Current Liabilities	Ideal: 2:1. Measures short-term liquidity.
Quick Ratio (Acid Test)	$(\text{Current Assets} - \text{Inventory} - \text{Prepaid}) / \text{Current Liabilities}$	Ideal: 1:1. Stricter liquidity test.
Absolute Liquid Ratio	$(\text{Cash} + \text{Bank} + \text{Short-term Investments}) / \text{Current Liabilities}$	Ideal: 0.5:1. Most conservative measure.
Debt-Equity Ratio	Long-term Debt / Shareholders' Equity	Ideal: 2:1. Measures leverage. Lower = safer.
Proprietary Ratio	Shareholders' Funds / Total Assets	Share of owners in total assets. Higher = safer.
Debt to Total Assets Ratio	Total Debt / Total Assets	Lower = lower financial risk.
Interest Coverage Ratio	EBIT / Interest Charges	Ideal: >3. Ability to pay interest from earnings.
Capital Gearing Ratio	Fixed Interest Bearing Securities / Equity Shareholders' Funds	High gearing = high risk & high return potential.

= Solved Example – Liquidity Ratios

Given: Current Assets = Rs.4,00,000 | Inventory = Rs.1,20,000 | Prepaid = Rs.20,000 | Cash & Bank = Rs.80,000 | Current Liabilities = Rs.2,00,000 | Long-term Debt = Rs.3,00,000 | Equity = Rs.5,00,000

Ratio	Calculation	Result
Current Ratio	$4,00,000 / 2,00,000$	= 2:1 ✓ (Ideal)
Quick Ratio	$(4,00,000 - 1,20,000 - 20,000) / 2,00,000$	= 2,60,000 / 2,00,000 = 1.3:1 ✓
Absolute Liquid Ratio	$80,000 / 2,00,000$	= 0.4:1 (Slightly below ideal)

Ratio	Calculation	Result
Debt-Equity Ratio	3,00,000 / 5,00,000	= 0.6:1 (Good – Low risk)

■ **EXAM TIP: Remember ideal ratios: Current = 2:1 | Quick = 1:1 | Absolute = 0.5:1 | D/E = 2:1. These appear frequently in exams.**

2.5 FUND FLOW STATEMENT

Concept of 'Fund': Fund means **Working Capital** (Current Assets – Current Liabilities). Any change that affects working capital is a 'flow of funds'.

Fund Flow Statement: It is a statement showing the **sources and uses (applications) of funds** during an accounting period. It explains why the working capital changed between two balance sheet dates.

Steps to Prepare Fund Flow Statement:

- Step 1: Calculate Working Capital at Beginning and End of the period
- Step 2: Calculate Change in Working Capital (Increase/Decrease)
- Step 3: Prepare Adjusted P&L; Account to find Funds from Operations
- Step 4: Identify Sources of Funds (long-term items that increase funds)
- Step 5: Identify Applications/Uses of Funds (items that decrease funds)
- Step 6: Prepare the Fund Flow Statement (Sources = Applications ± Change in WC)

Sources of Funds vs Applications of Funds

Sources of Funds	Applications of Funds
Issue of shares or debentures	Redemption of preference shares / debentures
Long-term borrowings (loans)	Repayment of long-term loans
Sale of fixed assets	Purchase of fixed assets
Funds from operations (Net Profit + Non-cash charges)	Loss from operations
Decrease in working capital	Increase in working capital
Capital introduced by proprietor	Drawings by proprietor / Dividend paid

— Solved Example – Fund Flow Statement

Given Data (Extract from Balance Sheets):

Particulars	2022 (Rs.)	2023 (Rs.)
Share Capital	5,00,000	6,00,000
Reserves & Surplus	1,00,000	1,50,000
Long-term Loan	2,00,000	3,00,000
Fixed Assets (Net)	6,00,000	7,50,000

Particulars	2022 (Rs.)	2023 (Rs.)
Current Assets	3,50,000	4,20,000
Current Liabilities	1,50,000	1,70,000

Solution:

Working Capital	2022 (Rs.)	2023 (Rs.)
Current Assets	3,50,000	4,20,000
Less: Current Liabilities	1,50,000	1,70,000
Net Working Capital	2,00,000	2,50,000
Increase in Working Capital	—	50,000

SOURCES	Amount (Rs.)
Issue of Share Capital (6,00,000 – 5,00,000)	1,00,000
Long-term Loan taken (3,00,000 – 2,00,000)	1,00,000
Funds from Operations (Reserves increase + Depreciation assumed Rs. 30,000)	
Total Sources (A)	2,80,000
APPLICATIONS	
Purchase of Fixed Assets (7,50,000 – 6,00,000 + 30,000 dep.)	1,80,000
Increase in Working Capital	50,000
Total Applications (B)	2,30,000

2.6 CASH FLOW STATEMENT (AS-3)

Meaning: A Cash Flow Statement shows **inflows and outflows of cash and cash equivalents** during an accounting period. It is governed by **Accounting Standard AS-3** in India.

Three Activities under Cash Flow Statement:

Activity	Description
Operating Activities	Core business activities – receipts from customers, payments to suppliers, employees, taxes. Most important section.
Investing Activities	Purchase/sale of long-term assets (machinery, investments). Represents growth investments.
Financing Activities	Issue/redemption of shares/debentures, repayment of loans, payment of dividends.

■ **NOTE:** Cash Flow from Operating Activities can be computed using (1) Direct Method – actual cash receipts/payments, or (2) Indirect Method – starting from Net Profit and adjusting non-cash items. Indirect Method is most common in practice.

— Solved Example – Cash Flow Statement (Indirect Method)

Given: Net Profit = Rs.2,00,000 | Depreciation = Rs.50,000 | Increase in Debtors = Rs.30,000 | Decrease in Creditors = Rs.20,000 | Purchase of Machinery = Rs.3,00,000 | Issue of Shares = Rs.2,50,000 | Dividend Paid = Rs.40,000 | Opening Cash = Rs.80,000

A. CASH FLOW FROM OPERATING ACTIVITIES	Rs.
Net Profit before tax	2,00,000
Add: Depreciation (non-cash)	50,000
Less: Increase in Debtors	(30,000)
Less: Decrease in Creditors	(20,000)
Net Cash from Operating Activities (A)	2,00,000
B. CASH FLOW FROM INVESTING ACTIVITIES	
Purchase of Machinery	(3,00,000)
Net Cash used in Investing Activities (B)	(3,00,000)
C. CASH FLOW FROM FINANCING ACTIVITIES	
Issue of Shares	2,50,000
Dividend Paid	(40,000)
Net Cash from Financing Activities (C)	2,10,000
Net Increase in Cash (A+B+C)	1,10,000
Add: Opening Cash Balance	80,000
Closing Cash Balance	1,90,000

■ **EXAM TIP:** Key difference: Fund Flow Statement deals with WORKING CAPITAL. Cash Flow Statement deals with CASH only. AS-3 governs Cash Flow Statement.

UNIT III — BUSINESS BUDGETING, MARGINAL COSTING & BREAK EVEN ANALYSIS

3.1 BUSINESS BUDGETING

Budget: A budget is a **formal quantitative statement** prepared in advance, expressing management's plans for a specified future period in financial and/or non-financial terms.

Budgeting: Budgeting is the process of preparing budgets. It is the act of **formulating, executing, and controlling** a budget.

Budgetary Control: A system where actual performance is compared with budgeted performance and corrective action is taken. It combines planning and control.

Objectives of Budgeting:

- Planning and coordination of activities
- Control of expenditure and costs
- Performance measurement of departments
- Profit maximisation through efficient resource allocation
- Motivation of employees by setting targets
- Facilitating communication across levels of management

Limitations of Budgeting:

- Based on estimates – may not be accurate
- Time-consuming and expensive to prepare
- May create rigidity in management thinking
- May lead to inter-departmental conflicts over resource allocation
- Frequent changes in business conditions make budgets obsolete quickly
- Success depends on top management support and employee cooperation

Essentials of Effective Budgeting:

- Clear organisational goals and objectives
- Strong support from top management
- Participation of all departments (participative budgeting)
- Realistic and attainable targets
- Proper communication of budget to all concerned
- Regular monitoring and review of actual vs budgeted performance
- Flexibility to adjust to changing conditions

(A) Flexible Budget

Meaning: A Flexible Budget is one that is designed to change with the **level of activity (output)**. It provides different cost targets for different output levels and is more realistic than a fixed budget.

Features: It separates costs into Fixed, Variable, and Semi-variable. It can be re-cast at any activity level. It is useful in situations where output fluctuates significantly.

— Flexible Budget Example

Cost Item	Budget at 80% Capacity	Budget at 100% Capacity	Budget at 120% Capacity
Production (Units)	8,000	10,000	12,000
Variable Costs:			
Direct Material (Rs.10/unit)	80,000	1,00,000	1,20,000
Direct Labour (Rs.8/unit)	64,000	80,000	96,000
Variable Overhead (Rs.4/unit)	32,000	40,000	48,000
Total Variable Cost	1,76,000	2,20,000	2,64,000
Fixed Costs:			
Factory Overhead	50,000	50,000	50,000
Admin Overhead	30,000	30,000	30,000
Total Fixed Cost	80,000	80,000	80,000
TOTAL COST	2,56,000	3,00,000	3,44,000
Cost Per Unit	Rs.32.00	Rs.30.00	Rs.28.67

(B) Zero Based Budgeting (ZBB)

Meaning: Zero Based Budgeting starts from **ZERO** each budget period. Every expenditure must be freshly justified regardless of previous budgets. There is no 'carry forward' from last year.

Advantages: Eliminates wasteful spending | Forces departments to justify all costs | Encourages innovative thinking | Allocates resources to high-priority activities.

Disadvantages: Time-consuming | Requires extensive documentation | May create anxiety among managers | Not suitable for all types of organisations.

■ **EXAM TIP: ZBB vs Traditional Budgeting: Traditional = last year + increment. ZBB = start from scratch each year. ZBB is more thorough but more expensive.**

3.2 MARGINAL COSTING

Meaning: Marginal Costing is a costing technique where **only variable costs** are charged to products. Fixed costs are treated as period costs and written off to Profit & Loss Account in the period incurred.

Key Concepts in Marginal Costing

Marginal Cost = Variable Cost per Unit (Direct Material + Direct Labour + Variable Overhead)

Contribution = Sales – Variable Cost (or Selling Price – Marginal Cost)

Profit = Contribution – Fixed Cost

P/V Ratio (Profit Volume Ratio) = (Contribution / Sales) × 100

P/V Ratio = (Change in Profit / Change in Sales) × 100

Determination of Profit under Marginal Costing:

Particulars	Per Unit (Rs.)	Total (Rs.)
Sales	XX	XX
Less: Variable Cost (Marginal Cost)	XX	XX
Contribution (Sales – Variable Cost)	XX	XX
Less: Fixed Cost	—	XX
Profit	—	XX

— Solved Example – Marginal Costing and Profit

Given: Selling Price = Rs.50/unit | Variable Cost = Rs.30/unit | Fixed Cost = Rs.80,000 | Production & Sales = 10,000 units

Particulars	Working	Amount (Rs.)
Sales (10,000 × Rs.50)		5,00,000
Less: Variable Cost (10,000 × Rs.30)		(3,00,000)
Contribution		2,00,000
Less: Fixed Cost		(80,000)
Profit		1,20,000
P/V Ratio	2,00,000/5,00,000 × 100	= 40%

Make or Buy Decision:

A firm should **MAKE** a product if the **marginal cost of making is less than the purchase price** (buying cost). Fixed costs are irrelevant as they are incurred either way.

— Make or Buy Decision Example

Given: Purchase Price = Rs.40/unit | Cost to Make: Direct Material Rs.12, Direct Labour Rs.8, Variable Overhead Rs.6, Fixed Overhead Rs.10 (unavoidable)

Cost to Make (relevant)	Rs./unit
Direct Material	12
Direct Labour	8
Variable Overhead	6
Total Marginal Cost to Make	26
Purchase Price	40

Decision: MAKE (saving Rs.14/unit)

3.3 BREAK EVEN ANALYSIS (BEP)

Concept: Break Even Point is the level of output/sales at which **Total Revenue = Total Cost** — there is neither profit nor loss. It is also called the 'no-profit-no-loss point'.

All Important BEP Formulas

BEP (in Units) = Fixed Cost / Contribution per Unit

BEP (in Sales Rs.) = Fixed Cost / P/V Ratio

P/V Ratio = Contribution / Sales (or) (SP – VC) / SP

Contribution per Unit = Selling Price per Unit – Variable Cost per Unit

Margin of Safety (Units) = Actual Sales – BEP Sales

Margin of Safety (Rs.) = (Actual Profit / P/V Ratio)

Margin of Safety Ratio = Margin of Safety / Actual Sales × 100

Profit = Margin of Safety × P/V Ratio

Required Sales for Target Profit = (Fixed Cost + Target Profit) / P/V Ratio

Composite BEP (Multiple Products) = Total Fixed Cost / Weighted Average P/V Ratio

— Comprehensive BEP Solved Example

Given: Selling Price = Rs.100/unit | Variable Cost = Rs.60/unit | Fixed Cost = Rs.2,00,000 | Actual Sales = 8,000 units

Contribution/unit	Rs.100 – Rs.60 = Rs.40
P/V Ratio	40/100 × 100 = 40%
BEP (Units)	2,00,000 / 40 = 5,000 units
BEP (Sales Rs.)	2,00,000 / 0.40 = Rs.5,00,000
Actual Sales Value	8,000 × Rs.100 = Rs.8,00,000
Margin of Safety (Units)	8,000 – 5,000 = 3,000 units
Margin of Safety (Rs.)	3,000 × Rs.100 = Rs.3,00,000
Margin of Safety Ratio	3,00,000 / 8,00,000 × 100 = 37.5%
Actual Profit	3,000 × Rs.40 = Rs.1,20,000 (or Rs.3,00,000 × 40% = Rs.1,20,000)
Sales for Target Profit Rs.1,60,000	(2,00,000 + 1,60,000) / 0.40 = Rs.9,00,000

■ EXAM TIP: BEP formula: $\text{Fixed Cost} \div \text{Contribution per Unit}$ (for units) OR $\text{Fixed Cost} \div \text{P/V Ratio}$ (for rupees). Margin of Safety = how far actual sales are ABOVE breakeven – wider is safer!

UNIT IV — STANDARD COSTING, VARIANCE ANALYSIS & REPORTING TO MANAGEMENT

4.1 STANDARD COSTING

Meaning: Standard Costing is a technique where **pre-determined costs (standards)** are set for products/services. Actual costs are compared with standard costs, and the difference (**Variance**) is identified and analysed.

Objectives of Standard Costing:

- To provide a basis for cost control by comparing actual vs standard
- To help in pricing decisions by knowing standard product cost
- To measure efficiency of production and workforce
- To simplify cost accounting records
- To motivate employees by setting performance targets
- To assist in budgeting and planning

Setting of Standards:

Type	How Set
Material Standard	Standard price per unit of material × Standard quantity required per unit of output
Labour Standard	Standard rate per hour × Standard hours required per unit of output
Overhead Standard	Budgeted Overhead / Budgeted Output (or Hours)

4.2 MATERIAL VARIANCE – COMPLETE ANALYSIS

Material Variance analyses the difference between **Standard Material Cost** and **Actual Material Cost** for actual output.

Material Variance Formulas

Standard Material Cost (SMC) = Std Qty for Actual Output × Std Price

Actual Material Cost (AMC) = Actual Qty × Actual Price

MCV (Material Cost Variance) = SMC – AMC [+ve = Favourable, –ve = Adverse]

MPV (Material Price Variance) = Actual Qty × (Std Price – Actual Price)

MUV (Material Usage Variance) = Std Price × (Std Qty for Act. Output – Actual Qty)

MMV (Material Mix Variance) = Std Price × (Std Mix of Actual Input – Actual Mix)

MYV (Material Yield Variance) = Std Cost per unit × (Actual Yield – Std Yield for Actual Input)

VERIFICATION: MCV = MPV + MUV | MUV = MMV + MYV

– Comprehensive Material Variance – Solved Example

Given: Actual Output = 900 units | Standard: 1 kg per unit @ Rs.20/kg | Actual: 1,000 kg used @ Rs.18/kg | Standard Mix = 60% Mat A + 40% Mat B

Variance	Formula Applied	Calculation	Result
Std Material Cost	$900 \times 1 \text{ kg} \times \text{Rs.20}$	$900 \text{ kg} \times \text{Rs.20}$	Rs.18,000
Actual Material Cost	$1,000 \text{ kg} \times \text{Rs.18}$	$1,000 \times 18$	Rs.18,000
MCV	$\text{SMC} - \text{AMC}$	$18,000 - 18,000$	Rs.0 (Nil)
MPV	$\text{AQ} \times (\text{SP} - \text{AP})$	$1,000 \times (20 - 18)$	Rs.2,000 (F)
MUV	$\text{SP} \times (\text{SQ} - \text{AQ})$	$\text{Rs.20} \times (900 - 1,000)$	Rs.(2,000) (A)
Verification	$\text{MPV} + \text{MUV}$	$2,000 + (-2,000)$	$= 0 = \text{MCV} \checkmark$

■ **NOTE: F = Favourable (Actual cost < Standard cost – GOOD). A = Adverse/Unfavourable (Actual cost > Standard cost – BAD). Always state F or A after each variance.**

4.3 LABOUR VARIANCE – COMPLETE ANALYSIS

Labour Variance analyses the difference between **Standard Labour Cost** and **Actual Labour Cost** for actual output.

Labour Variance Formulas

Standard Labour Cost (SLC) = Std Hours for Actual Output × Std Rate

Actual Labour Cost (ALC) = Actual Hours Worked × Actual Rate

LCV (Labour Cost Variance) = SLC – ALC [+ve = F, –ve = A]

LRV (Labour Rate Variance) = Actual Hours × (Std Rate – Actual Rate)

LEV (Labour Efficiency Variance) = Std Rate × (Std Hours for Act. Output – Actual Hours Worked)

LTV (Labour Time Variance) = Std Rate × (Std Hours – Actual Hours Paid for)

Labour Idle Time Variance = Std Rate × Idle Hours (always Adverse)

VERIFICATION: LCV = LRV + LEV

= Comprehensive Labour Variance – Solved Example

Given: Actual Output = 500 units | Std: 2 hours per unit @ Rs.15/hour | Actual: 1,100 hours worked @ Rs.16/hour | Idle Time = 50 hours

Variance	Formula	Calculation	Result
Std Labour Cost	$500 \times 2 \times \text{Rs.15}$	$1,000 \text{ hrs} \times \text{Rs.15}$	Rs.15,000
Actual Labour Cost	$1,100 \times \text{Rs.16}$	$1,100 \times 16$	Rs.17,600
LCV	$\text{SLC} - \text{ALC}$	$15,000 - 17,600$	Rs.2,600 (A)
LRV	$\text{AH} \times (\text{SR} - \text{AR})$	$1,100 \times (15 - 16)$	Rs.1,100 (A)
LEV	$\text{SR} \times (\text{SH} - \text{AH})$	$\text{Rs.15} \times (1,000 - 1,100)$	Rs.1,500 (A)
Idle Time Var.	$\text{SR} \times \text{Idle Hrs}$	$\text{Rs.15} \times 50 \text{ hrs}$	Rs.750 (A)
Verification	$\text{LRV} + \text{LEV}$	$1,100 + 1,500$	$= 2,600 = \text{LCV} \checkmark$

■ **EXAM TIP:** Master the verification: $\text{MCV} = \text{MPV} + \text{MUV}$. $\text{LCV} = \text{LRV} + \text{LEV}$. In every exam question, **VERIFY** your answer at the end to score full marks.

4.4 REPORTING TO MANAGEMENT

Meaning: Reporting to Management involves the **preparation and presentation of financial and non-financial information** to management at different levels to facilitate planning, control, and decision-making.

Objectives of Management Reporting:

- Provide relevant and timely information for decision-making
- Help management evaluate performance against plans/budgets
- Identify problems and variances requiring attention
- Facilitate coordination between different departments
- Serve as a basis for future planning
- Ensure accountability at all levels of management

Principles of Good Management Reporting:

Principle	Explanation
Relevance	Reports must contain information relevant to the decision at hand.
Accuracy	Data must be correct and verifiable; errors reduce trust.
Timeliness	Reports must reach managers in time to take action.
Clarity	Must be clear, concise, and easily understood by the recipient.
Comparability	Should enable comparison with previous periods or budgets.
Consistency	Same format and basis should be used over time.

Principle	Explanation
Economy	Cost of preparing report should not exceed its benefits.
Exception Principle	Reports should highlight significant deviations only.

Classification of Reports:

Basis	Types
By Time	Daily Reports, Weekly Reports, Monthly Reports, Annual Reports
By Function	Production Reports, Sales Reports, Financial Reports, HR Reports, Purchase Reports
By Nature	Routine Reports (regular, periodic), Special Reports (one-time for specific purpose)
By Form	Written Reports, Oral Reports, Graphic Reports (charts, graphs)
By Level	Top Management Reports, Middle Management Reports, Operational Reports
By Purpose	Information Reports (just data), Analytical Reports (data + analysis), Recommendation Reports

Reporting at Different Levels of Management:

Top Management (Board / CEO)	Annual financial statements, profitability summaries, capital budgeting reports, strategic
Middle Management (Departmental)	Budget vs actual variance reports, departmental cost reports, productivity reports, sales
Lower Management (Supervisor)	Daily production reports, labour efficiency records, material usage reports, scrap & waste

■ **EXAM TIP: Reporting question: Always write Meaning + Objectives + Principles + Classification + Levels. This covers the full 10-mark answer structure.**

QUICK REFERENCE — ALL IMPORTANT FORMULAS

PROFITABILITY RATIOS

Gross Profit Ratio = $GP / \text{Net Sales} \times 100$

Net Profit Ratio = $\text{Net Profit} / \text{Net Sales} \times 100$

ROCE = $\text{Net Profit} / \text{Capital Employed} \times 100$

ROE = $\text{Net Profit} / \text{Shareholders Equity} \times 100$

EPS = $\text{Net Profit after tax} / \text{No. of Equity Shares}$

LIQUIDITY & SOLVENCY RATIOS

Current Ratio = $\text{Current Assets} / \text{Current Liabilities}$ [Ideal: 2:1]

Quick Ratio = $(CA - \text{Stock} - \text{Prepaid}) / CL$ [Ideal: 1:1]

Debt-Equity Ratio = $\text{Long-term Debt} / \text{Equity}$ [Ideal: 2:1]

Interest Coverage = $EBIT / \text{Interest Charges}$ [Ideal: >3]

BEP & MARGINAL COSTING FORMULAS

Contribution = $\text{Sales} - \text{Variable Cost}$

P/V Ratio = $\text{Contribution} / \text{Sales} \times 100$

BEP (Units) = $\text{Fixed Cost} / \text{Contribution per Unit}$

BEP (Rs.) = $\text{Fixed Cost} / \text{P/V Ratio}$

Margin of Safety = $\text{Actual Sales} - \text{BEP Sales}$

Profit = $\text{Margin of Safety} \times \text{P/V Ratio}$

Sales for Target Profit = $(\text{Fixed Cost} + \text{Target Profit}) / \text{P/V Ratio}$

VARIANCE ANALYSIS FORMULAS

MCV = $\text{Std Cost} - \text{Actual Cost}$ | $MCV = MPV + MUV$

MPV = $AQ \times (SP - AP)$ | $MUV = SP \times (SQ - AQ)$

MMV = $SP \times (\text{Std Mix of Actual} - \text{Actual Mix})$

$MYV = \text{Std Cost/unit} \times (\text{Actual Yield} - \text{Std Yield for Actual Input})$

$LCV = SLC - ALC \mid LCV = LRV + LEV$

$LRV = AH \times (SR - AR) \mid LEV = SR \times (SH - AH)$

$\text{Idle Time Variance} = SR \times \text{Idle Hours}$ [Always Adverse]